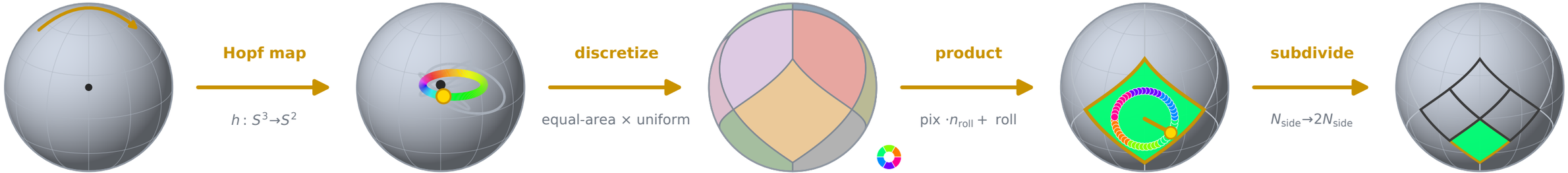
The Hopf--HEALPix discretization of $SO(3)$: one equivolumetric partition by construction



$SO(3)$

rotations,
Haar volume π^2

$S^2 \times S^1$

the actual Hopf fibre
(roll ψ along it)

HEALPix × roll

$N_{\text{pix}} = 12N_{\text{side}}^2$ equal-area
$\times\, N_\psi = 6N_{\text{side}}$ roll bins

one cell

equivolumetric:
$\mu(\text{cell}) = \pi^2 / N(\ell)$

refine ×8

pixel ×4, roll ×2
$N(\ell) = 72 \cdot 8^\ell$